

LESSON GUIDE

Technology Plus

Unit 7: Security

6.4.2 – Data Types

Lesson Overview:

In this lesson, students will explore common uses of encryption and differentiate between data in transit and data at rest.

Students will:

- Identify and describe common uses of encryption for data in transit and data at rest.
- Differentiate between data in transit and data at rest.
- Analyze a scenario and explain how encryption protects the confidentiality and integrity of data

Guiding Question: How does encryption protect data both when it is stored and when it is moving across networks?

Suggested Grade Levels: 8-12

Technology Needed: No

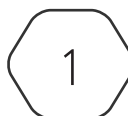
Approximate Time Required: 65-85 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.4 – Identify common use cases for encryption.
 - Plain text vs. cipher text
 - **Data at rest**
 - **File level**
 - **Disk level**
 - **Mobile device**
 - **Data in transit**
 - **Email**
 - **HTTPS**
 - **VPN**
 - **Mobile application**

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Lesson Background

Background Information:

In today's digital world, almost everything we do involves data. We store personal files on laptops and phones, send private messages to friends, shop online, and even access our grades through the internet. This information is valuable – and if it falls into the wrong hands, it could be stolen, read, or even changed.

Encryption is a way to protect that data. It scrambles information into unreadable code that can only be unlocked with the right key. This means that even if a malicious actor, thief, or stranger gets access, they can't understand or misuse the data. There are two main times we need encryption: data at rest, when information is stored on a device or server (like files, photos, or backups); data in transit, when information is moving across a network (like emails, messages, or online purchases).

Understanding these two states of data helps us see why encryption is used everywhere, from securing smartphones to protecting bank accounts and private conversations.

Students will learn more about file encryption in lessons 1.2.3 and 3.1.1, data persistence in lesson 5.2.1, and privacy in lesson 6.1.4.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.4.2 – Data Types • Activity 6.4.2 – Destination Encryption
Materials per Student	• Student Notes 6.4.2 – Data Types • Student Handout 6.4.2 – Data Types • Assessment 6.4.2 – Data Types (optional)

Cyber Terms:

Term	Definition
encryption	the process of converting information or data into an unreadable code, specifically to prevent unauthorized access
data at rest	data that is stored physically in any digital form that is not actively moving through networks or being processed
file level encryption	encryption applied directly to individual files or folders
disk level encryption	encryption applied to an entire disk or storage device, including all files and folders on it
mobile device encryption	encryption applied specifically to data stored on mobile devices such as smartphones and tablets
data in transit	data actively moving from one location to another, such as across networks or between systems
email encryption	ensures that only the intended recipients can read the message
Hyper Transfer Protocol Secure (HTTPS)	an extension of HTTP that uses TLS/SSL (Transport Layer Security/Secure Socket Layer) to encrypt the data transmitted between a web browser and a web server
Virtual Private Network (VPN)	creates a secure, encrypted tunnel between the user's device and the VPN server, masking the user's IP address and protecting data from being intercepted by third parties
mobile application encryption	often used to secure data transmitted between the app and its servers

6.4.2 – Data Types

Guiding Question: How does encryption protect data both when it is stored and when it is moving across networks?

Lesson Launch (10 minutes)

- Imagine you lost your phone in the cafeteria. It has your texts, photos, saved passwords, even banking info. What could happen if the person who finds it could open it?
- Give students time to think and allow a few to share their thoughts.

Students should conclude that the person could see their private information or even try to steal their identity.
- Now imagine you're on the school Wi-Fi and send a private message to a friend. What if someone nearby could read it as easily as a postcard without an envelope?
- Give students time to think and allow a few to share their thoughts.

Students should conclude that their message would no longer be private, and someone could learn personal details about them.
- Ask:** Would you feel safe if anyone could see your messages, grades, social media, or payment apps?
- Pause and allow students to answer. Accept all reasonable answers.
- Transition:** Tell students that today they will learn how encryption keeps data private and secure – whether it's stored on a device or sent across the internet

Encryption Everywhere (25-35 minutes)

- Remind students about **encryption**:
 - Any data can be encrypted, but not all data needs to be.
 - Example: a social media post vs. your bank details.
 - Data exists in two states: At Rest and In Transit.
 - Data should be encrypted when it is private, personal, involves money, could be stolen, is sent online, is on public Wi-Fi, or is sensitive to others.
 - Rule of thumb: If you wouldn't want a stranger to read, steal, or change it, encrypt it.

Data At Rest

- Definition: stored data not actively moving (files on computers, stored emails, backups, etc.).
- File-level encryption** → Ideal for protecting specific sensitive files or folders.
- Disk-level encryption** → Ideal for securing an entire device, like a laptop or external drive. Protects data even if the device is lost, but still requires authentication.
- Mobile device encryption** → Ideal for smartphones/tablets that are easy to lose or steal. Secured

with PINs, passwords, or biometrics.

Data In Transit

- Definition: data actively moving (emails, web browsing, chats, calls, streaming, VPNs).
- **Email encryption** → Ideal for sensitive info shared via email. End-to-end keeps it secure all the way; TLS protects only while in transmission.
- **HTTPS** → Ideal for online banking, shopping, and login pages. Look for the padlock in the browser.
- **VPNs** → Ideal for remote access and public Wi-Fi. Common protocols like OpenVPN, IPsec, and L2TP encrypt traffic.
- **Mobile application encryption** → Ideal for securing app-based communication and financial transactions. Uses TLS/SSL in transit and application-level encryption before data leaves the device.

Activity: Destination Encryption (20-30 minutes)

- Display Activity Slides 6.4.2 – Data Types and have students record their answers in the handout.
- Decide if students will work independently or in small groups.
- Read each scenario aloud (or allow students/groups to read independently).
- For each scenario, students will determine the state of the data, the type of encryption being used, and how encryption helps protect it.
- Allow time for students to write their answers, then call on a few students/groups to share before moving on.
- If time is short, select only a few scenarios to discuss as a class.
- Wrap up by reinforcing how the scenarios illustrate data at rest and data in transit.

Activity Handout

Directions: Read each scenario carefully. Determine whether the data is at rest or in transit, the type of encryption being used, and briefly explain how encryption is helping protect the data. Record your responses below.

Scenario	Data State	Encryption Type	How Encryption Helps
A student keeps their laptop in their backpack every day at school. One day, the laptop was stolen, but the thief cannot read the files without the right password.	Data at Rest	Disk-level encryption	Protects all files on the laptop if someone gains unauthorized access
A shopper enters personal details and payment information on an online checkout page. The browser shows a small padlock symbol near the website address.	Data in Transit	HTTPS encryption	Protects personal and payment info during online purchases
An employee works from home but needs to connect to their company's private network. They use special software that creates a secure "tunnel" to send and receive data safely.	Data in Transit	VPN encryption	Creates a secure tunnel for remote work traffic
A person drops their phone while getting off the bus. Although the phone is found by someone else, the finder cannot see the personal photos, apps, or contacts.	Data at Rest	Mobile device encryption	Prevents access to photos, apps, and contacts if the device is lost

A college student emails an application with sensitive personal information to a potential employer. The email cannot be read by anyone intercepting it along the way.	Data in Transit	Email Encryption	Keeps sensitive info in email safe from interception
A photographer saves a collection of private images on an external USB drive. If someone were to take the drive, they would not be able to open the files without permission.	Data at Rest	File-level encryption	Keeps images safe even if USB is stolen
A person transfers money to a friend using a mobile banking app. The details of the transaction cannot be read by anyone else on the internet.	Data in Transit	Mobile application encryption	Ensures financial transactions over the internet through an app are secure

Putting It All Together (10 minutes)

- Review what students learned today by discussing the guiding question:
- How do the imaginary scenarios from the lesson launch relate to today's lesson?
 - Lost phone scenario → data at rest, protected with mobile device encryption.
 - Wi-Fi message scenario → data in transit, protected with tools like VPNs or mobile app encryption.
- **Ask:** When should data be encrypted?

Data should be encrypted when it is private, has personal info, involves money, could be stolen, is sent over the internet, is on public Wi-Fi, or is sensitive to others. A simple rule: if you wouldn't want a stranger to read, steal, or change it, encrypt it.
- **Ask:** What is the difference between data at rest and data in transit?

Data at rest = stored physically in digital form.

Data in transit = actively moving between locations or systems.
- **Ask:** How can we protect data at rest?

With file-level encryption, disk-level encryption, or mobile device encryption.
- **Ask:** How can we protect data in transit?

With email encryption, HTTPS, VPNs, or mobile application encryption.
- Close by reminding students: Encryption keeps data private and secure, no matter where it is.